

Assemble your printed bench mechanism



YOUR TWO PRINT BATCHES

This guide uses exactly the **28 printed pieces** from the **25 STL files** in your two batches, plus one M3 x16 screw and one M3 nut.

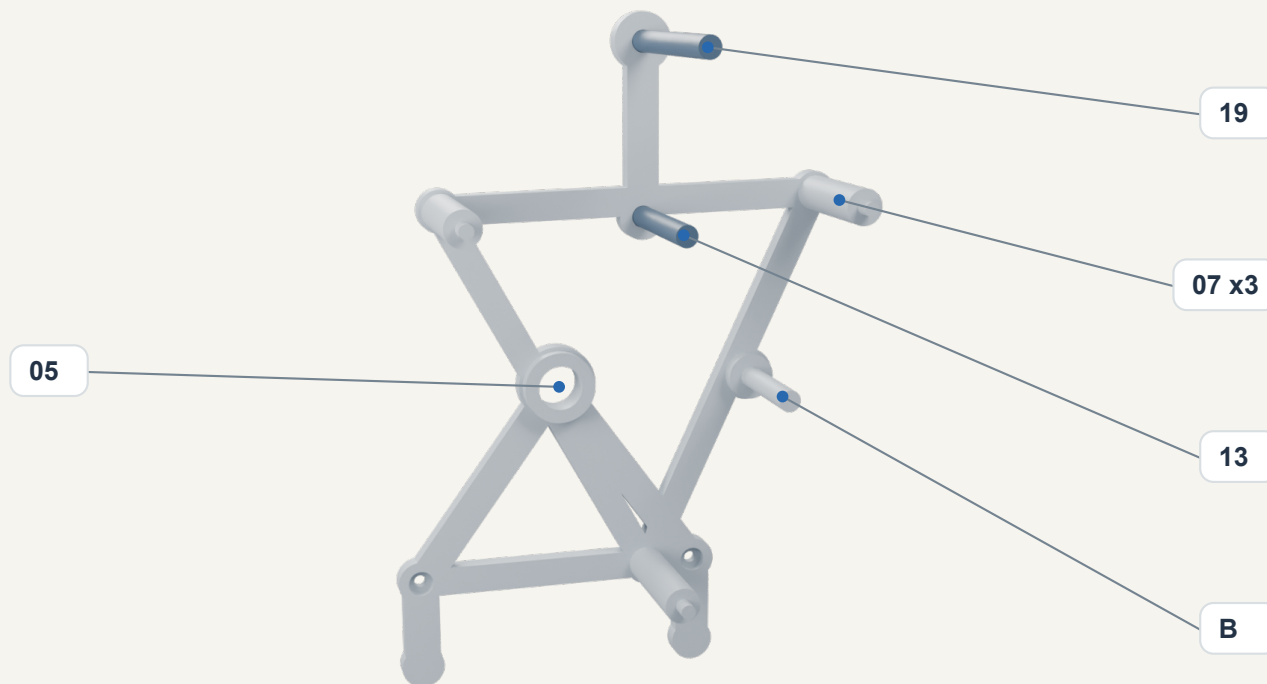
The result is a supported bench mechanism: gear train, escapement and pendulum. You supply a gentle turning force by hand for this first test.

Front means the pendulum side, facing you in these pictures. **Rear** means the side with the heads of pivots 13 and 19.

- Frames, posts and bridge
- 10 - orange input wheel
- 11 - green compound
- 12 - yellow escape wheel
- 18 / 24 / 25 - regulator
- Printed pins and spacers

Before assembly: remove brims and obvious seam burrs. Try shafts in their bores by hand. Clean tight shaft surfaces gently; preserve gear teeth and pallet working faces. Keep the bob empty until the mechanism is upright.

Build the rear support



PARTS TO ADD NOW

05_rear_carrier.stl
 07_carrier_post.stl x3
 13_gear_pivot.stl
 19_anchor_pivot.stl

1. Hold rear frame **05** with its long built-in B pin pointing toward you. The two small bowl screw holes are at the bottom; leave them empty.
2. Insert short pivot **13** through hole C and long pivot **19** through the top hole P, both **from behind**. Their flat heads stay behind frame 05.
3. Insert the three posts **07** into the three small corner holes shown. Their thick shoulders sit against the gear-side face of 05.

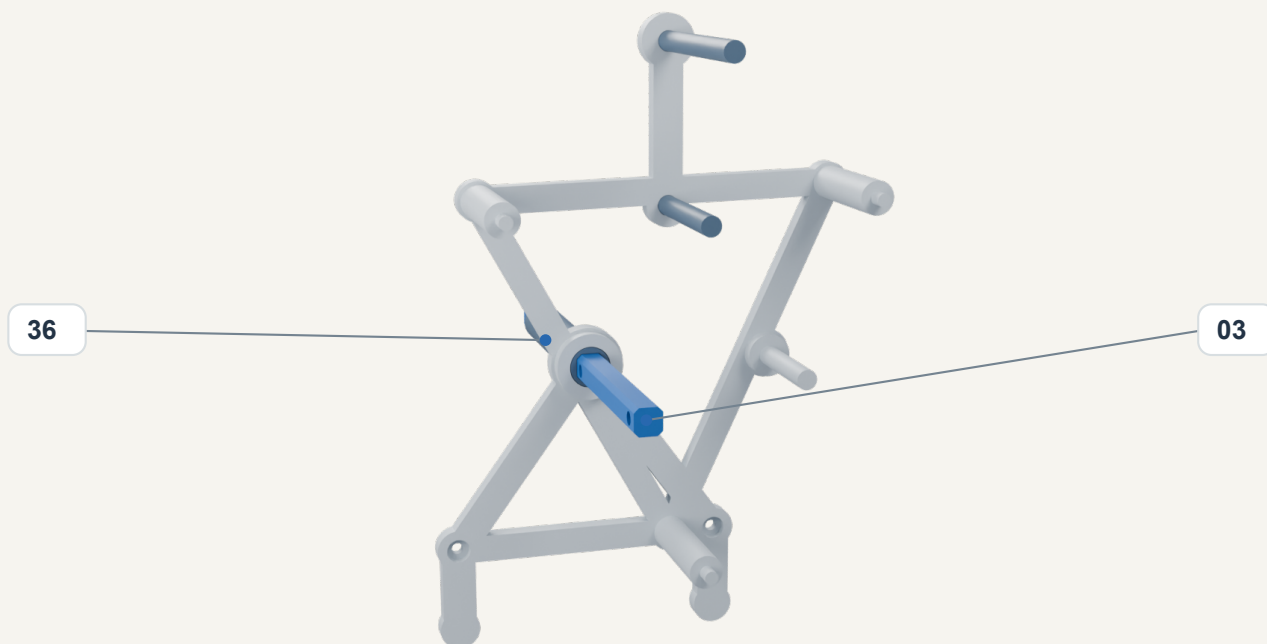
Looking from the gear / pendulum side. B is already printed into frame 05.

FOUR WORKING AXES

A: large main bearing. B: built-in gear pin. C: escape-wheel pivot 13. P: top anchor pivot 19. The three separate corner holes take posts 07; none goes into a bowl screw hole.

Check: 13 is 49 mm overall; 19 is 67 mm. Use all three 47 mm carrier posts. Set front frame 06 aside.

Fit the square shaft and rear sleeve



PARTS TO ADD NOW

03_main_shaft.stl

36_rear_round_journal.stl

1. Identify the **rear end** of shaft **03**: the middle cross-hole is **81 mm** from it, and 59 mm from the other end. Mark the rear end temporarily.
2. Slide long round sleeve **36** onto the square shaft. Feed the sleeved shaft through the large A hole of frame 05 from behind.
3. Set the front end of sleeve 36 **flush with the front face of the raised collar on 05**. Hold it there while adding the next parts.

The long round sleeve passes through the rear frame bearing.

SHAFT 03

140 mm overall

5 mm

81 mm

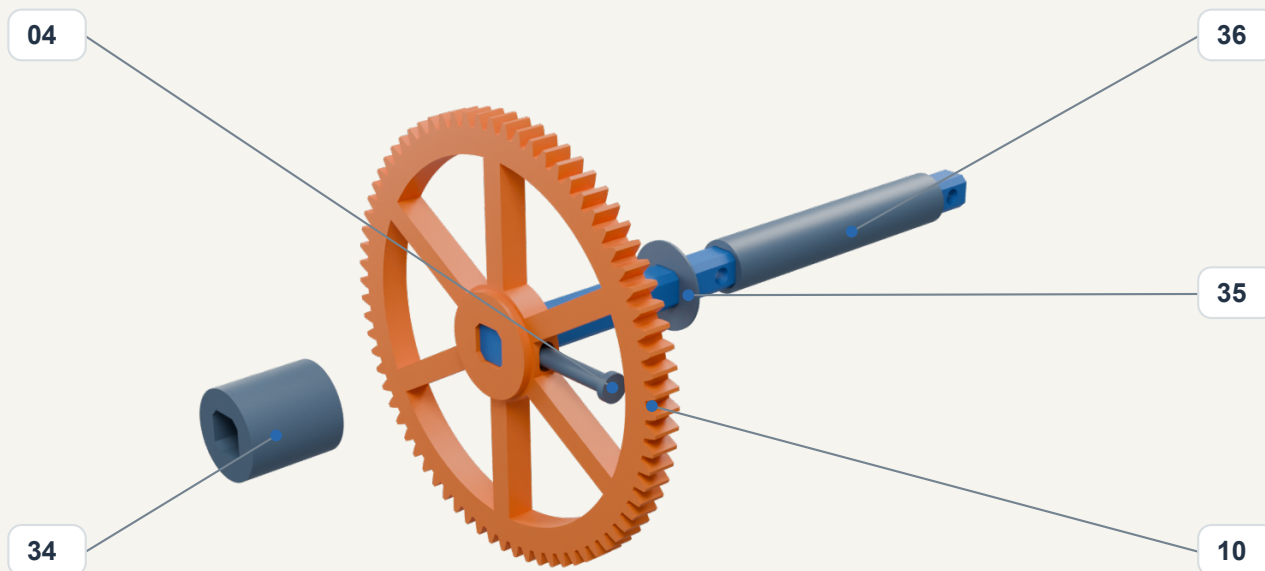
135 mm

REAR END

FRONT END

Check: the 66.6 mm sleeve provides the round bearing surface. The bare square shaft should not rub directly in the frame hole.

Add the orange input gear



PARTS TO ADD NOW

35_main_rear_thrust_washer.stl

10_drive_72.stl

04_main_crosspin.stl

34_main_front_thrust_sleeve.stl

1. From the front, slide thin washer **35** onto the square shaft, against the rear collar. This is the very thin **0.6 mm** washer.

2. Slide drive gear **10** on with its **taller central hub facing forward**. Align its cross-slot with the shaft hole 81 mm from the rear end.

3. Push crosspin **04** through the hub and shaft. Then add the larger keyed sleeve **34** in front of the gear. Its length is **19.2 mm**.

Exploded detail: parts are spread apart along the shaft. Frame omitted here.

REAR > FRONT

05

rear collar

>

35

0.6 mm washer

>

10 + 04

gear + pin

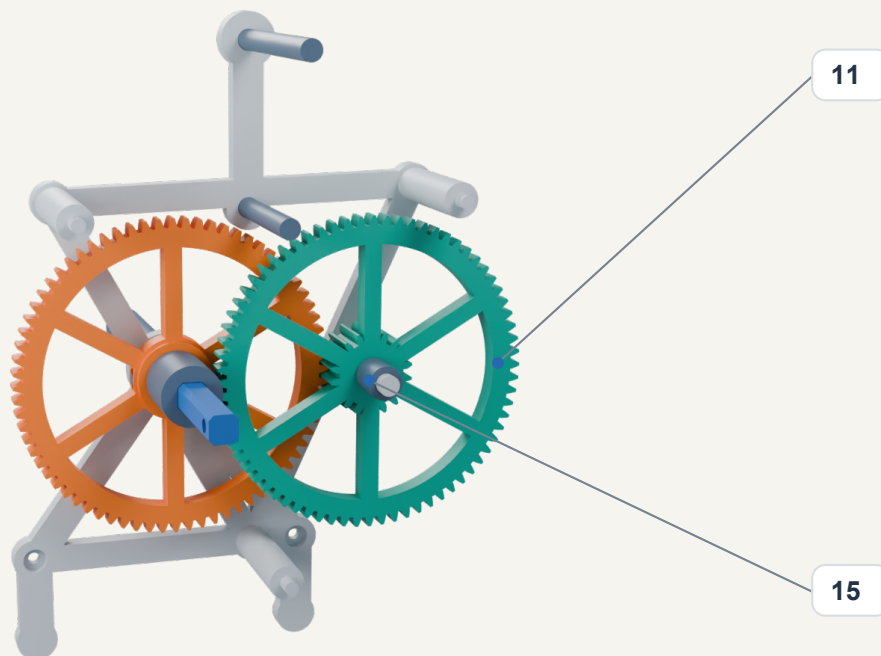
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34

19.2 mm sleeve

Check: gear 10 and shaft 03 turn together. The square socket transmits torque; a little tangential play in the retaining pin slot is intentional.

Fit the green compound gear



PARTS TO ADD NOW

11_compound_18_72.stl

15_B_front_spacer.stl

1. Slide compound gear **11** onto the built-in B pivot of rear frame 05.
2. Point its **projecting small pinion toward the rear frame**. Its large 72-tooth wheel faces forward. The small pinion meshes with orange gear 10.
3. Slide spacer **15**, length **11.4 mm**, onto the B pivot in front of gear 11.

The green wheel sits farther forward than the orange wheel.

REAR > FRONT

05 / B

integral pivot

>

11

pinion rearward

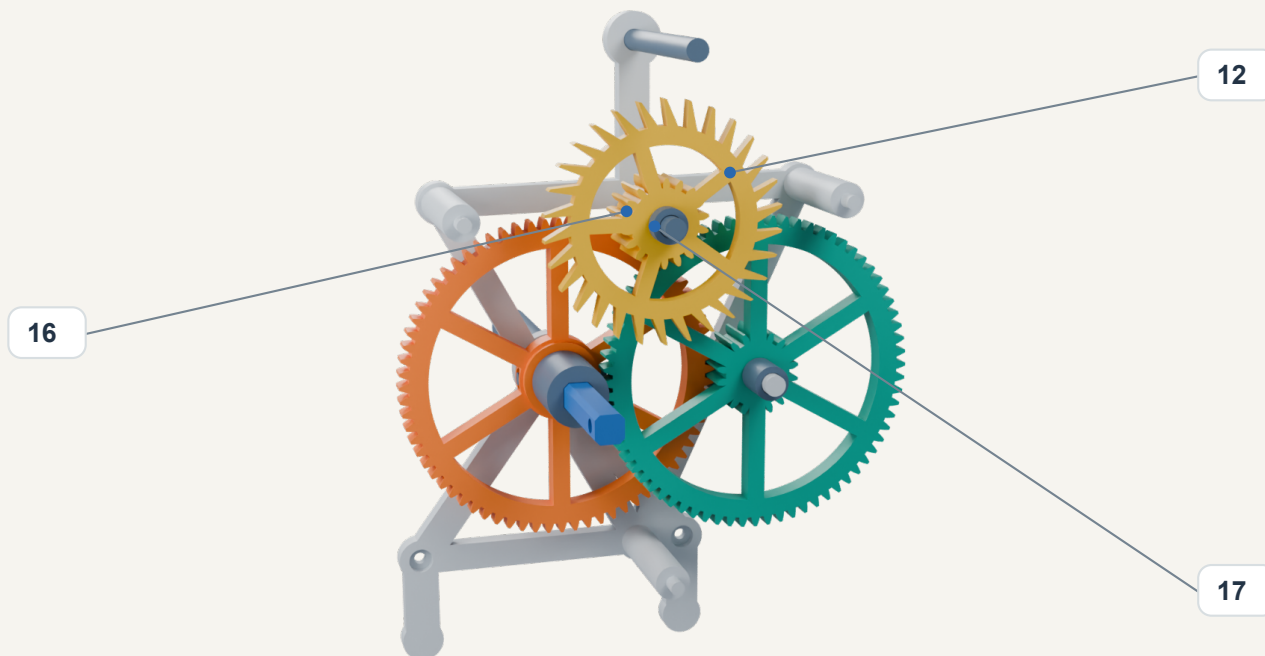
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15

11.4 mm spacer

Check: turn the gears gently while supporting the loose shafts. The large green wheel should clear the orange wheel rather than mesh with it.

Fit the yellow escape compound



PARTS TO ADD NOW

16_C_rear_spacer.stl
12_escape_compound.stl
17_C_front_spacer.stl

1. On pivot **13** at C, slide on rear spacer **16**, length **14.4 mm**. It rests against frame 05.

2. Add escape compound **12**: its projecting small pinion points **rearward**, and its large pointed escape wheel faces **forward**. Turn gently to engage the green wheel.

3. Add front spacer **17**, length **4.4 mm**. Before fitting the anchor, check that both gear meshes turn without a hard spot.

The yellow small pinion engages the large green wheel.

REAR > FRONT

05

rear frame

>

16

14.4 mm

>

12

pinion rearward

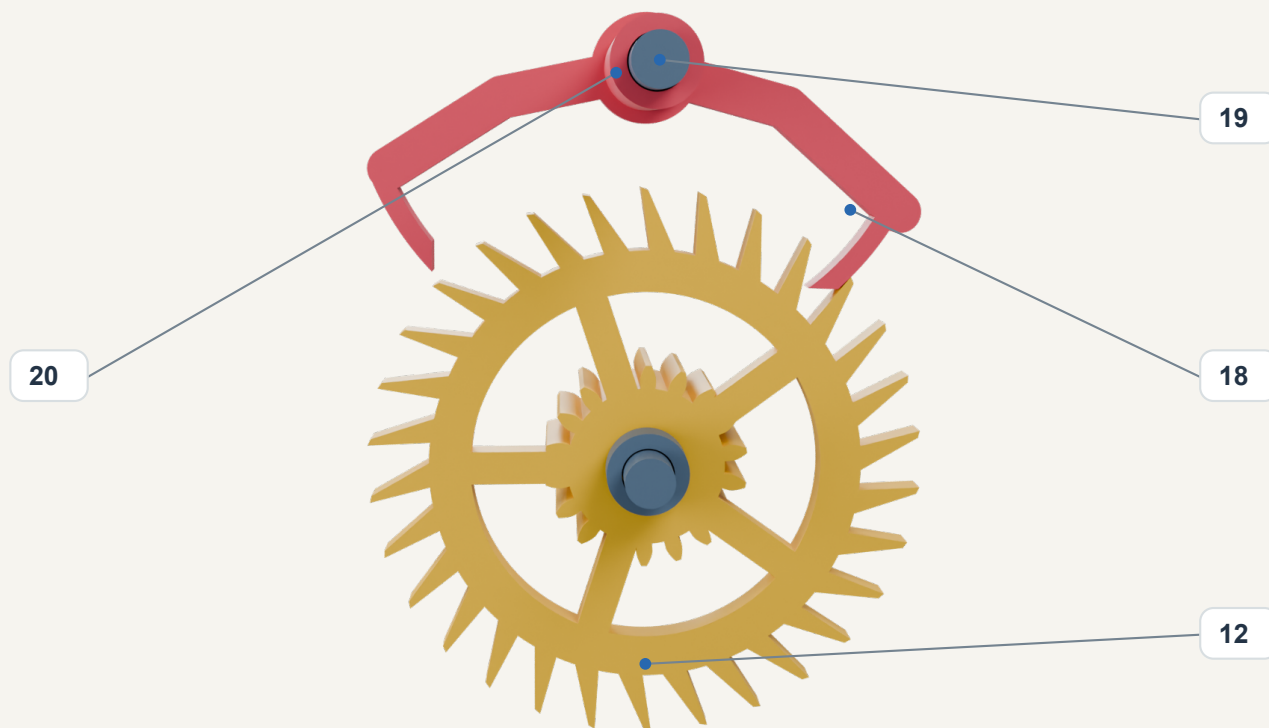
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17

4.4 mm

Check: 16 is the long spacer behind the yellow gear; 17 is the short spacer in front. Do not press on the pointed tooth tips.

Place the pallet anchor



PARTS TO ADD NOW

20_anchor_rear_spacer.stl

18_pallet_anchor.stl

1. On long top pivot **19**, slide rear spacer **20** against frame 05. This narrow spacer is **27.4 mm** long.

2. Slide anchor **18** onto that pivot with its two pallet arms pointing down toward the escape wheel. Its long round sleeve points **forward**.

3. Ease the pallets around the upper teeth of wheel 12, rotating the wheel a little if necessary. The pallet faces and escape wheel occupy the **same front-to-back plane**.

Isolated escapement detail. Rear frame and other gears omitted for clarity.

REAR > FRONT

05

rear frame

>

20

27.4 mm spacer

>

18

arms at wheel

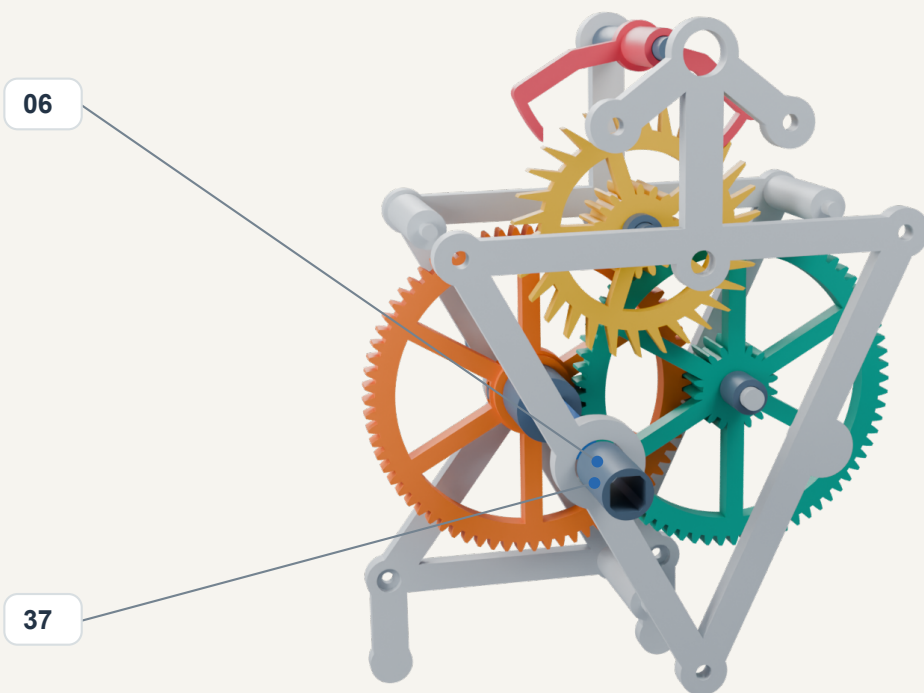
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sleeve

points forward

Check: the anchor pivots freely on 19. Preserve the little working faces and tooth tips; do not force the wheel past a locked pallet.

Close the two frames



PARTS TO ADD NOW

06_front_carrier.stl

37_front_round_journal.stl

1. Orient front frame **06** with its raised main collar and **blind B bearing seat facing rearward**, toward the gears.
2. Bring it onto all three posts 07. Guide B's built-in pin into its blind seat, C pivot 13 into its hole, and the anchor's round sleeve through the large top opening.
3. Seat the frame gently. Slide short round sleeve **37** over the front of the square shaft and into the A bearing. Its rear end should be **flush with the rear face of the front frame collar**.

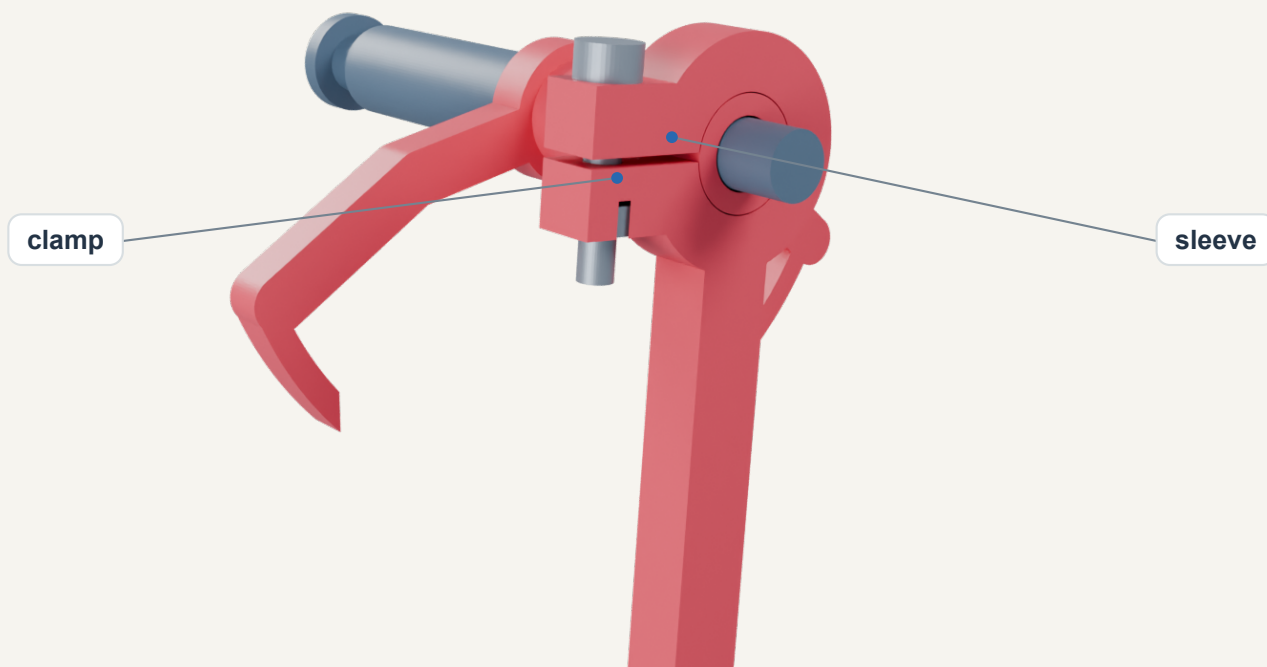
Front frame and front sleeve are shown pulled forward before seating.

SEAT THE FRAME EVENLY

Posts locate the two frames. If a joint is loose during the bench test, a small removable tape wrap around the fixed joint can hold it; keep tape away from teeth and rotating surfaces. Do not glue the assembly yet.

Check: sleeves 36 and 37 turn with the shaft. Keep a little axial play in the gears. If closure makes anything stiff, reopen and check the spacer order.

Clamp the pendulum to the anchor



Close-up of the anchor sleeve and split pendulum clamp. Rod continues below.

LEAVE ACCESS FOR BEAT ADJUSTMENT

The angle between anchor and pendulum is adjustable. Tightening the clamp makes them move together; it must not pinch the stationary pivot. The illustrated screw and nut are unthreaded visual envelopes.

Check: the pendulum clamps to the rotating anchor sleeve, not to the stationary 6 mm pivot. Keep the clamp clear of frame 06.

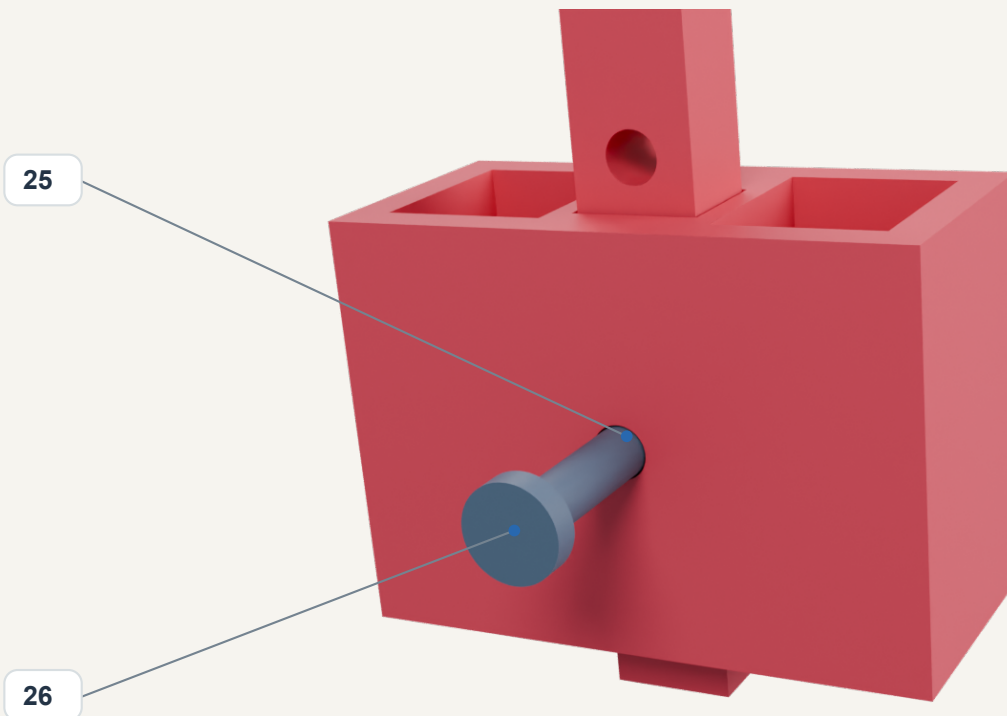
PARTS TO ADD NOW

24_pendulum.stl

+ M3 x16 screw and plain M3 nut

1. Slide the split clamp of pendulum **24** over the anchor's protruding round sleeve, ahead of front frame 06. The rod points downward.
2. Place one plain **M3 nut** in the hexagonal recess of the clamp ear. Insert an **M3 x16 screw** through the opposite ear and start it into the nut.
3. Set the front face of the clamp approximately flush with the end of the anchor sleeve. Tighten only enough to hold it provisionally; you will adjust its angle during the test.

Fit the open bob



PARTS TO ADD NOW

25_open_pendulum_bob.stl

26_bob_pin.stl

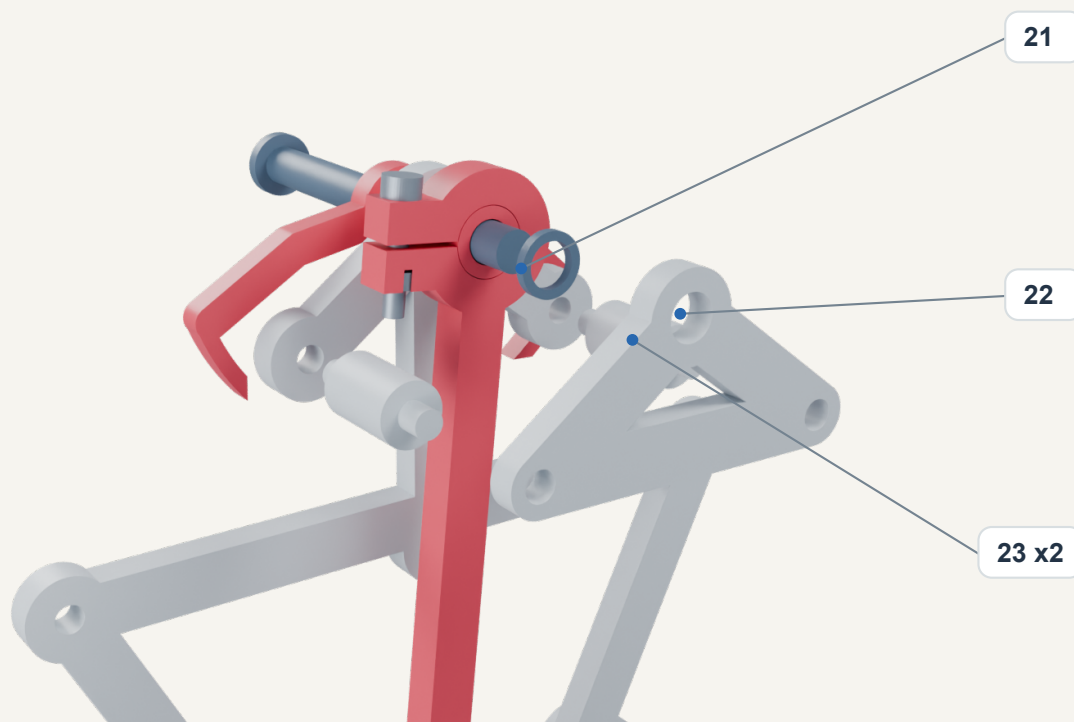
1. Slide open bob **25** onto the bottom of the pendulum rod with its two open pockets facing **upward**.
2. Align its hole with the **150 mm** rod hole: the **fourth hole from the pivot**, or the **second hole from the free end**.
3. Insert bob pin **26**. Keep the bob empty until the assembled frame is supported upright; then put **15-20 g total** of iron into its two pockets.

Rear close-up of the bob; its pin is pulled back before insertion.



Check: the bob pin passes through the rod and bob. Split the iron approximately evenly between the two pockets; this is the only ballast used in this bench test.

Support the front of the anchor pivot



Exploded close-up: washer, two bridge posts and bridge are pulled forward.

REAR > FRONT

18 + 24

sleeve + clamp

>

21

1.4 mm washer

>

22

front bridge

PARTS TO ADD NOW

21_anchor_front_washer.stl

23_bridge_post.stl x2

22_anchor_bridge.stl

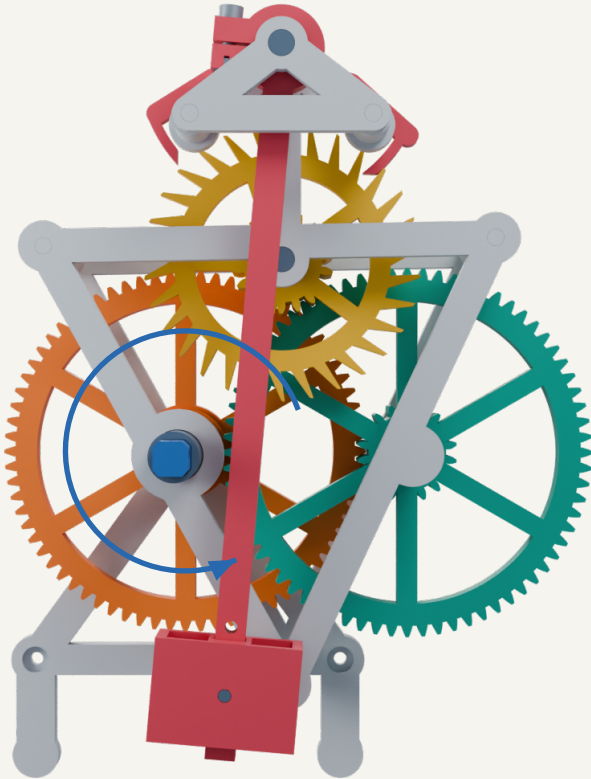
1. Slide thin washer **21**, thickness **1.4 mm**, over the exposed end of pivot 19, just in front of the pendulum clamp and anchor sleeve.

2. Fit both short bridge posts **23** into the two small holes beside the top of front frame 06. Each post is **23 mm overall**.

3. Fit bridge **22** onto the posts while guiding pivot 19 into its central hole. Seat all three locations together without bending the pivot.

Check: the anchor pivot is now supported at both ends. The anchor and pendulum must turn freely with slight endplay; the bridge must not squeeze the clamp.

Support it, then try the escapement



VIEWED FROM THE PENDULUM SIDE

Orange and yellow turn counterclockwise; green turns clockwise. This corrects the earlier clockwise instruction for this front view.

Pass: alternating catches without tooth skipping, scraping or a tight spot. If it binds or skips, stop and inspect the fit and pallet contact before adding force. This hand-driven check does not yet establish self-running performance.

1. Support the **fixed frame upright** with a padded clamp or blocks, leaving the pendulum and both gear faces free. Avoid squeezing the bearing areas.
2. Without drum wheels, sleeves **36 and 37 can slide outward**. Keep them seated. Small removable tape collars on the exposed shaft just outside their outer ends can retain them; keep tape away from the frame bearings.
3. Add **15-20 g** to the bob. Apply very light **counterclockwise torque to orange gear 10**, viewed from the pendulum side. Move the pendulum gently, initially about 5 degrees each side.
4. Look for **release, then a definite stop on the opposite pallet** on each half-swing. Adjust the pendulum clamp angle until release occurs at similar excursions either side of vertical; retighten lightly.

Keep these small parts straight

SPACERS AND SLEEVES

15	11.4 mm	In front of green gear 11
16	14.4 mm	Behind yellow gear 12
17	4.4 mm	In front of yellow gear 12
20	27.4 mm	Behind anchor 18
21	1.4 mm	Before the anchor bridge 22
34	19.2 mm	In front of orange gear 10
35	0.6 mm	Behind orange gear 10
36	66.6 mm	Long round sleeve; rear A bearing
37	23.6 mm	Short round sleeve; front A bearing

PINS, POSTS AND PIVOTS

07	3 copies	47 mm overall; joins frames
13	1 copy	49 mm overall; escape pivot
19	1 copy	67 mm overall; anchor pivot
23	2 copies	23 mm overall; bridge supports
04 / 26	1 each	Main gear retainer / bob retainer

IF SOMETHING DOES NOT FIT

Frame will not close

Check the three posts, the blind B seat and both pivot tips. Back off and align them together; do not use the front plate to bend a pin into place.

Gears stiff after closure

Check spacer lengths, shaft seams and remaining endplay. The front frame should not clamp a gear between two thrust faces.

Anchor does not swing freely

Check spacer 20, washer 21, bridge seating and clamp position. Nothing rotating should be trapped against a fixed frame or bridge.

Teeth pass without a firm catch

Confirm that the wheel and pallets are in the same plane, then adjust the beat. If catches remain marginal or skip, keep the tooth tips intact and record a close-up.

It stops when you release the gear

Expected for this bench setup: the rolling drum and its drive are not fitted. Keep very light hand torque while assessing the escapement.